"ARTIFICIAL VACUUM" FOR *T*-VIOLATION EXPERIMENT

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We point out the possible use of matrix isolation spectroscopy in experimental tests of fundamental physics. As a concrete example we show how this technique might be applied to measuring the *T*-violation electron electric dipole moment d_e. We estimate that a dipole moment could be detected down to $d_e \sim 6\times 10^{-28} e$ cm, in comparison to the current limit of $d_e < 10^{-24} e$ cm.

The problem of choosing a material with the necessary properties for an experimental test of fundamental physics can be complicated by the intricacies of the condensed phase. On the other hand the gas phase, while allowing the study of species with well-defined properties, must be at such a high temperature and low density that small signals would be swamped by thermal noise. We wish to point out the possibility of using matrix isolation to circumvent these difficulties.

Matrix isolation has been used for various forms of spectroscopy over the past 30 years [1,2]. To isolate atoms with a small amount of the element is evaporated, mixed with a noble gas and then deposited on a cryogenic plate. The matrix serves as an "artificial vacuum" in which the atoms remain a "gas" at a much higher density and lower temperature than possible in a real vacum.

As a concrete example of how matrix isolation might be applied, we would like to consider the problem of measuring the as yet undetected electric dipole moment (EDM) of the electron. The existence of an electron EDM is a possible result of an interaction which breaks time reversal invariance [3]. Evidence for such an interaction has thus far only been seen in kaon experiments, from which it has been impossible to determine the actual mechanism of *T*-violation. Detection of an electron EDM d_e would not only provide a second example of *T*-violation, but might also help determine the mechanism since different models predict different values for d_e. The current experimental limit is $d_e < 10^{-24}$ e cm [4].

Recently a new method was proposed for measuring d_e [5]. The proposed measurement consisted of using a superconducting quantum interference device (SQUID) to detect the magnetization induced in a paramagnetic material by an applied electric field.

The feasibility of the experiment depends on a combination of several material parameters (magnetic susceptibility and Curie temperatures, spin relaxation time, breakdown voltage). There seem to be few materials for which all the relevant parameters are documented. In ref. [5] EuS was considered as a possible paramagnetic material. We wish to consider matrix isolated Cs atoms instead.

There are two advantages to matrix isolated Cs atoms over EuS. First, the EDM measured is in fact that of the atom and is given by $d_{\text{eff}} = R d_e$ where R is an enhancement factor due to the mixing of atomic states of opposite parity [6]. For E_u^{++}, $R \approx 0.5$ [5]

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while for Cs, $R \approx 133$ [7]. Also, EuS has a low dielectric strength, breaking down with an applied field of $E \sim 10^3$ V/cm [8]. If this breakdown voltage applies at frequencies ~kHz, and cannot be improved by careful material processing, EuS will be of limited value. $E > 10^3$ V/cm is necessary to improve the present upper limit $d_e < 10^{-24}\, e$ cm [3].

In contrast, solid Ar does not break down until $E \gtrsim 10^5$ V/cm [9]. The Cs atom will, of course, affect the dielectric strength. We may estimate this by considering the breakdown due to the Cs atoms alone, ignoring the influence of the Ar matrix. Avalanche breakdown is due to the competing process of ionizing electron–atom scattering and non-ionizing inelastic scattering for which the cross sections are σ_{ion} and σ_{exc}, respectively. If one knows these cross sections one may estimate the dielectric strength of a material using [10]

$$E^* = \left(\ln \frac{L}{40 \tau_\mu E^*}\right)^{-1} \frac{m_e}{e} \int_0^{v_I} \frac{dv}{\tau(v)}, \qquad (1)$$

where E^* is the breakdown field, $\tau(v)$ is the mean time between inelastic (non-ionizing) collisions as a function of the electron's velocity v, v_I is the velocity of an electron with kinetic energy equal to the ionizing energy of the atom, L is the distance between the electrodes, τ is the mean free time for an electron at breakdown, and μ is the electron mobility at breakdown.

Using the cross sections for Cs [11] we estimate that a 0.1% concentration sample with a thickness of 5 cm will have a breakdown field of $E^* \approx 10^5$ V/cm. It should be noted that eq. (1) must be modified for Cs. Since for Cs $\sigma_{exc} \gg \sigma_{ion}$ dissipation will dominate over ionization even for energies above the ionization energy of Cs. Therefore, we replace v_I with the velocity at which σ_{ion} first reaches its peak value instead of the velocity at which σ_{ion} first becomes non-zero. Even at this velocity $\sigma_{ion} \approx 8 \times 10^{-16}$ cm^2 while $\sigma_{exc} \approx 10^{-14}$ cm^2 so our estimate appears to be a conservative one.

The cross section data shows that our assumption of treating Cs independently is valid since for Ar $\sigma_{exc}(E) \approx 0$ until $E \gtrsim 12$ eV [12], which is above the energy region of interest in calculating E^*_{Cs}. As a check we used the data in (12) to estimate $E^* \approx 2.5 \times 10^5$ V/cm for pure Ar with $L = 10^{-2}$ cm. This is in reasonable agreement with the measured value $E^* = 5.5 \times 10^5$ V/cm (10^{-2} cm thickness) [13].

Excess free electrons which can lead to breakdown can arise from injection at the cathode [14]. Therefore, for high dielectric strength one desires clean, smooth electrodes [15]. To ensure this the regions of the sample in contact with the electrodes should be pure Ar, eliminating any possible complications from Cs atoms interacting with the electrodes.

It should be noted that the choice of Ar is somewhat arbitrary. Another possible host is Kr, although it does perturb the ESR spectrum of Cs slightly more than Ar [16]. Xe would probably be a poor choice since the authors of ref. [16] were unable to detect an ESR signal from Cs in Xe.

There are two limits on the smallest detectable EDM, one from thermal fluctuations of the sample's magnetization and the other from noise associated with the SQUID. These limits are [4]

$$(d_{eff})^2_{TN} \sim (2.7 \times 10^{-29}\, e\,\text{cm})^2 \frac{T}{10\ \text{K}} \frac{\tau}{10^{-10}\ \text{s}} \times \frac{\Delta f}{10^{-4}\ \text{Hz}} \frac{10\ \text{cm}^3}{V} \frac{1}{\chi} \left(\frac{10^5\ \text{V/cm}}{E}\right)^2,$$

$$(d_{eff})^2_{SQUID} \sim (8.3 \times 10^{-28}\, e\,\text{cm})^2 \frac{\Delta f}{10^{-4}\ \text{Hz}} \times \frac{10\ \text{cm}^3}{V} \frac{1}{\chi^2} \left(\frac{10^5\ \text{V/cm}}{E}\right)^2,$$

where T is the temperature, τ is the spin lattice relaxation time, V is the sample volume, χ is the paramagnetic susceptibility, E is the applied electric field, and Δf is the bandwidth around the frequency at which E oscillates.

The material dependent quantities are τ and χ. First consider τ. Since $d_e < 10^{-24}\, e$cm [3], at an applied field of 10^5 V/cm the splitting between levels due to the EDM satisfies $\Delta E < 133 \times 10^{-19}$ eV. For any conceivable experiment we will therefore have $\Delta E \ll k_B T$, under which condition Cs will have a spin lattice relaxation time varying as [17]

$$\tau \propto T^{-9}.$$

Assuming that τ for Cs atoms in Ar is about the same as that for H atoms in Ar we may establish the con-

stant of proportionality with the experimental value $\tau \sim 10^{-3}$ s for H atoms in Ar at 4.2 K [18]

$$\tau \sim 2.5\times 10^{-3}\ \mathrm{K^9\ s}\ T^{-9}\ .$$

Next we consider the paramagnetic susceptibility χ. For $\Delta E \ll k_B T$ it is given by the Curie law [19]

$$\chi = \frac{N}{V}\frac{[g(JLS)\mu_B]^2}{3}\frac{J(J+1)}{k_B T}\ .$$

ESR experiments on various alkali metal atoms in rare gas matrices have been done with densities of $N/V = 10^{19}$ spins/cm^3, or $\sim 0.1\%$ metal atoms [20]. We adopt this as the density of Cs atoms, in which case $\chi = 6.2\times 10^{-6}\ \mathrm{K}/T$. With these expressions for χ and τ we find

$$(d_{\mathrm{eff}})^2_{\mathrm{TN}} \sim (1.7\times 10^{-23}\ e\,\mathrm{cm})^2 \frac{\Delta f}{10^{-4}\ \mathrm{Hz}}$$

$$\times \frac{10\ \mathrm{cm}^3}{V}\left(\frac{10^5\ \mathrm{V/cm}}{E}\right)^2\left(\frac{\mathrm{K}}{T}\right)^7\ ,$$

$$(d_{\mathrm{eff}})^2_{\mathrm{SQUID}} \sim (1.3\times 10^{-22}\ e\ \mathrm{cm})^2 \frac{\Delta f}{10^{-4}\ \mathrm{Hz}}$$

$$\times \frac{10\ \mathrm{cm}^3}{V}\left(\frac{10^5\ \mathrm{V/cm}}{E}\right)^2\left(\frac{T}{\mathrm{K}}\right)^2\ .$$

The optimal temperature at which to perform the experiment is $T_{\mathrm{opt}} = 0.64$ K, at which $(d_{\mathrm{eff}})_{\mathrm{SQUID}} = (d_{\mathrm{eff}})_{\mathrm{TN}}$. A standard experiment with $\Delta f = 10^{-4}$ Hz, $V = 10$ cm^3, and $E = 10^5$ V/cm could then have a noise limit of $d_{\mathrm{eff}} \sim 8.3\times 10^{-23}\ e$cm. With the enhancement factor for Cs we obtain for the electron $d_e \sim 6.3\times 10^{-25}\ e$cm, which is only slightly better than the current limit on d_e. However, the experiment may be scaled up. With a one liter sample and an integration time of $\sim$ one day ($\Delta f \sim 10^{-5}$ Hz) $d_e \sim 6.3\times 10^{-28} e$cm.

The manufacture of large samples merits discussion. The sample geometry should be such that thickness $\ll$ diameter. Therefore a one liter sample must be several cm thick. Fortunately the yield of isolated metal atoms increases with deposition rate, reaching a plateau of $\sim 0.1\%$ for a deposition rate $> 4\times 10^{-7}$ cm/s [2]. The deposition rate given by Jen et al. [12] for H atoms in Ar is 1.4×10^{-6} cm/s [18] which means the desired several cm thickness could be grown in several weeks. This should serve as an order of magnitude indication of the practicality of growing large samples.

Finally it should be noted that the possible dependence of χ and τ on the deposition procedure, and on the choice of T_{opt}, suggests that χ and τ should be carefully measured for the sample to determine the exact noise limit, and to determine T_{opt}. There is also the possibility of altering deposition conditions so as to optimize χ and τ.

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